

```
# =====  
# STAGE 7: DATA VISUALIZATION  
# =====  
  
# Import required libraries  
import pandas as pd  
import matplotlib.pyplot as plt  
  
# Load dataset  
df = pd.read_csv("upi_transactions_2024.csv")  
  
print("=" * 60)  
print("STAGE 7: DATA VISUALIZATION")  
print("=" * 60)  
  
print("\nRequired libraries imported:")  
print("1. Pandas")  
print("2. Matplotlib")  
  
print("\nDataset loaded successfully")  
print("Rows      :", df.shape[0])  
print("Columns   :", df.shape[1])
```

```
=====  
STAGE 7: DATA VISUALIZATION  
=====
```

Required libraries imported:

1. Pandas
2. Matplotlib

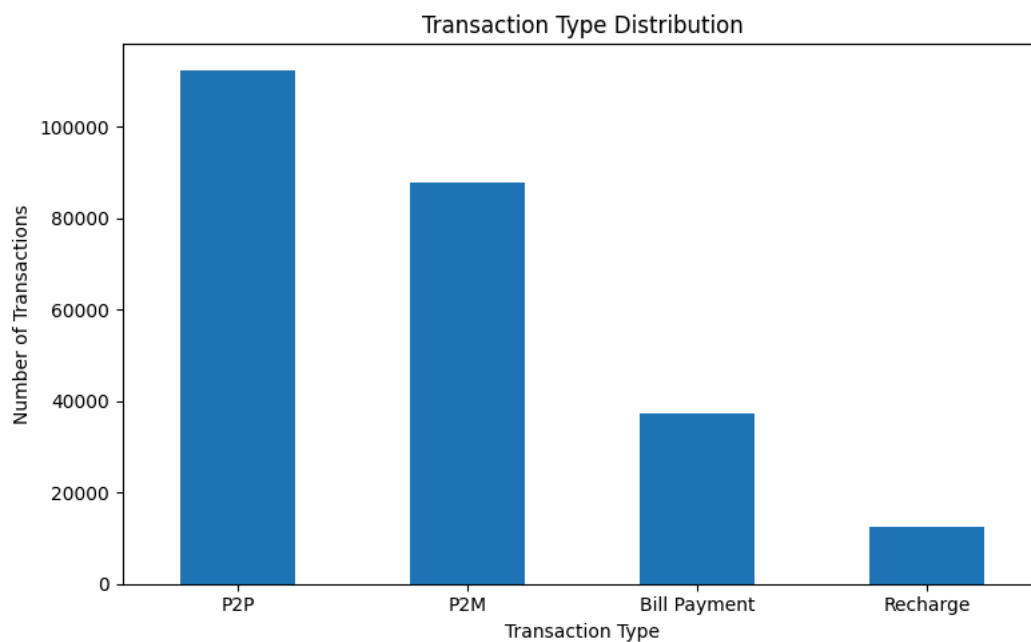
Dataset loaded successfully

Rows : 250000
Columns : 17

```
transaction_type_counts = df["transaction type"].value_counts()
```

```
plt.figure(figsize=(8, 5))  
transaction_type_counts.plot(kind="bar")
```

```
plt.title("Transaction Type Distribution")  
plt.xlabel("Transaction Type")  
plt.ylabel("Number of Transactions")  
plt.xticks(rotation=0)  
plt.tight_layout()  
plt.show()
```

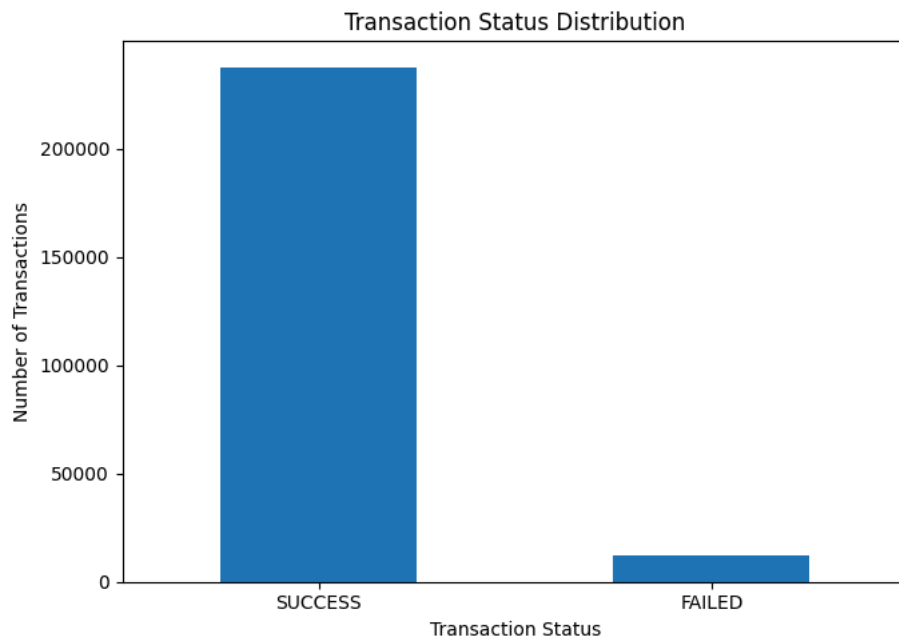


```
status_counts = df["transaction_status"].value_counts()
```

```
plt.figure(figsize=(7, 5))  
status_counts.plot(kind="bar")
```

```
plt.title("Transaction Status Distribution")
```

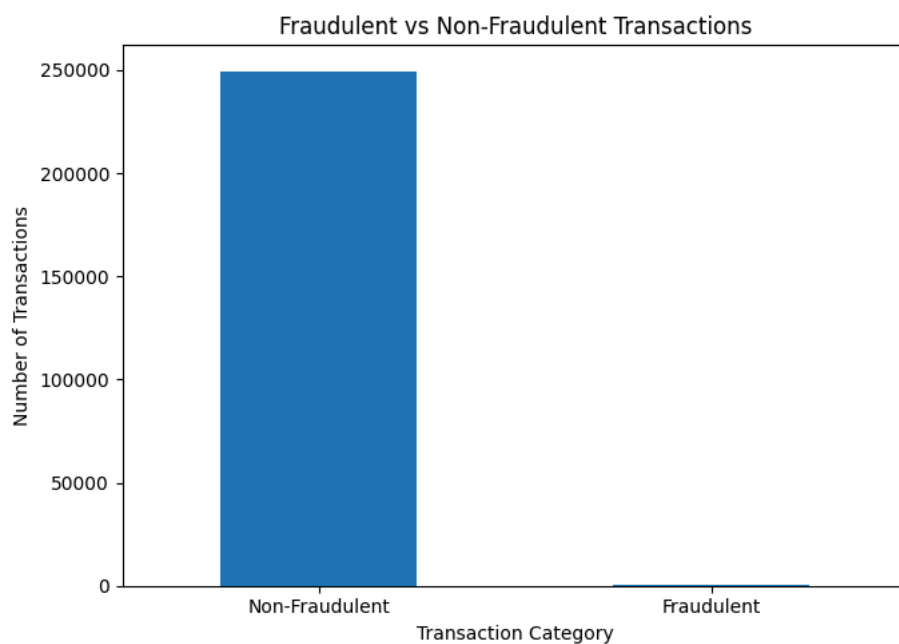
```
plt.xlabel("Transaction Status")
plt.ylabel("Number of Transactions")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



```
fraud_counts = df["fraud_flag"].map({
    0: "Non-Fraudulent",
    1: "Fraudulent"
}).value_counts()

plt.figure(figsize=(7, 5))
fraud_counts.plot(kind="bar")

plt.title("Fraudulent vs Non-Fraudulent Transactions")
plt.xlabel("Transaction Category")
plt.ylabel("Number of Transactions")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

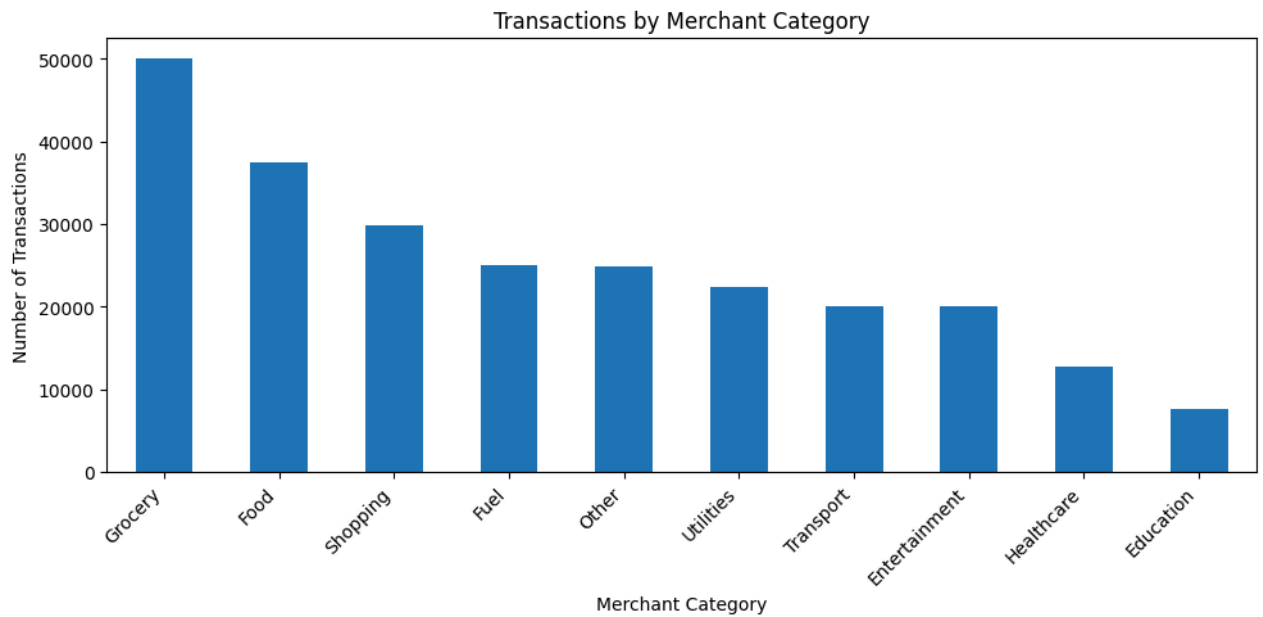


```
merchant_counts = df["merchant_category"].value_counts()

plt.figure(figsize=(10, 5))
merchant_counts.plot(kind="bar")

plt.title("Transactions by Merchant Category")
```

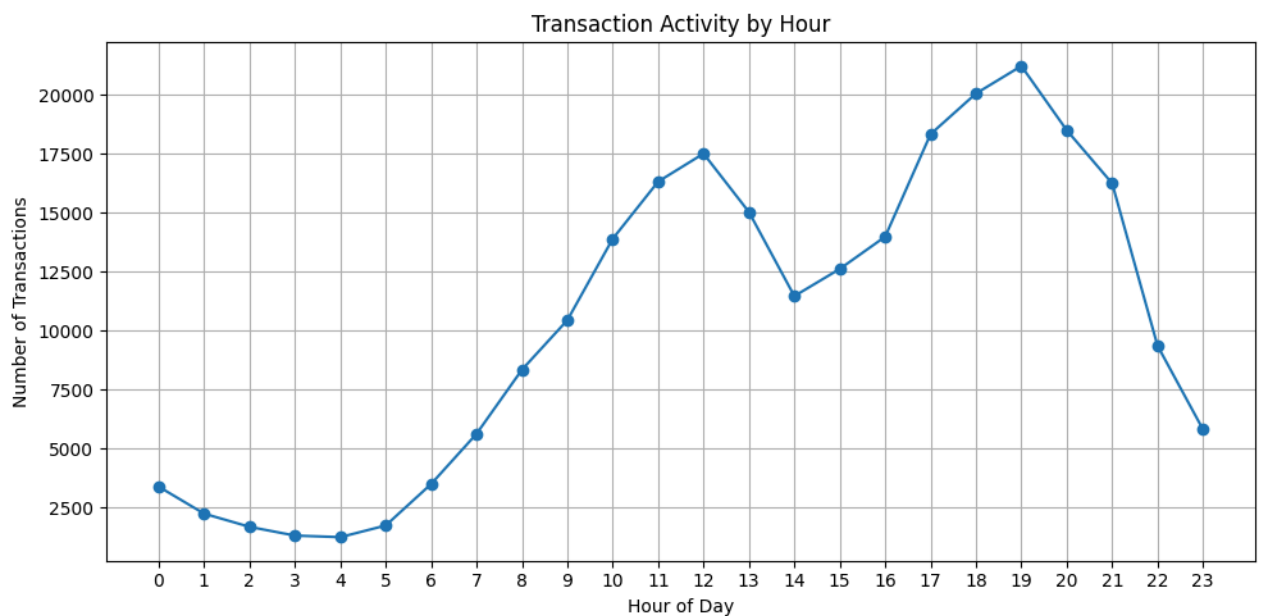
```
plt.xlabel("Merchant Category")
plt.ylabel("Number of Transactions")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```



```
hour_counts = df["hour_of_day"].value_counts().sort_index()
```

```
plt.figure(figsize=(10, 5))
hour_counts.plot(kind="line", marker="o")
```

```
plt.title("Transaction Activity by Hour")
plt.xlabel("Hour of Day")
plt.ylabel("Number of Transactions")
plt.xticks(range(24))
plt.grid(True)
plt.tight_layout()
plt.show()
```

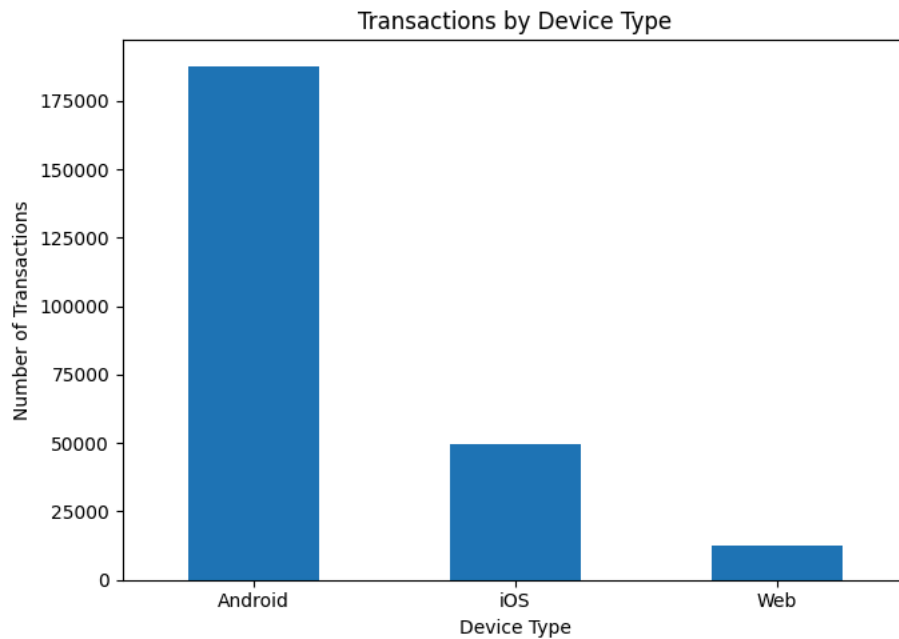


```
device_counts = df["device_type"].value_counts()
```

```
plt.figure(figsize=(7, 5))
device_counts.plot(kind="bar")
```

```
plt.title("Transactions by Device Type")
plt.xlabel("Device Type")
plt.ylabel("Number of Transactions")
```

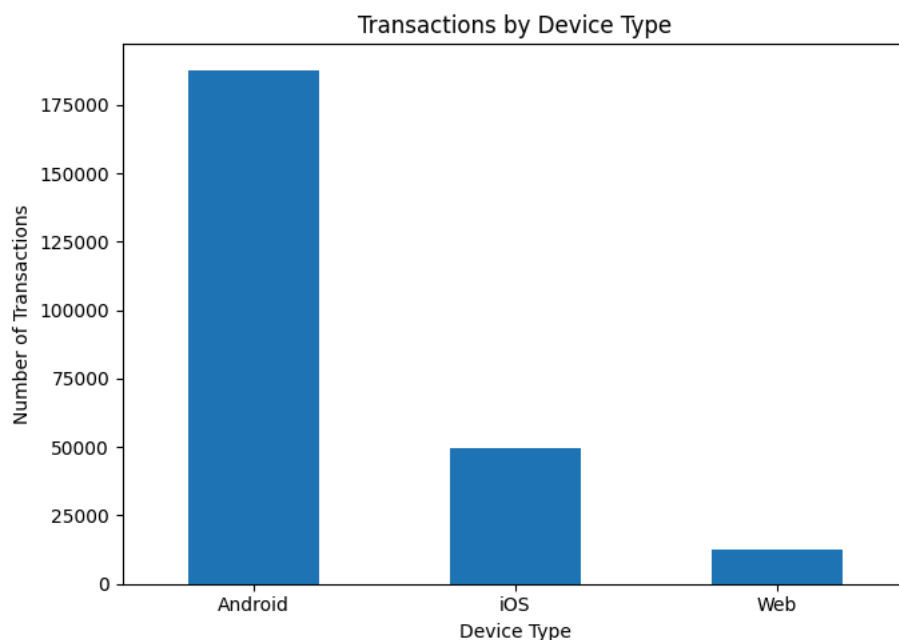
```
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



```
device_counts = df["device_type"].value_counts()

plt.figure(figsize=(7, 5))
device_counts.plot(kind="bar")

plt.title("Transactions by Device Type")
plt.xlabel("Device Type")
plt.ylabel("Number of Transactions")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

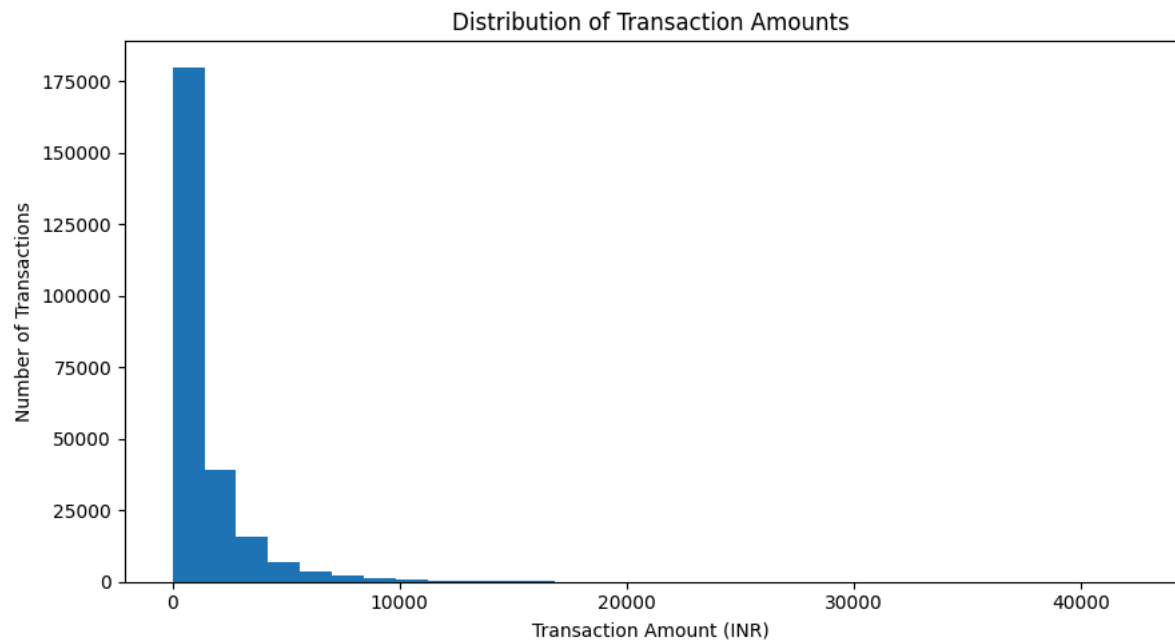


```
plt.figure(figsize=(9, 5))

plt.hist(
    df["amount (INR)"],
    bins=30
)

plt.title("Distribution of Transaction Amounts")
plt.xlabel("Transaction Amount (INR)")
plt.ylabel("Number of Transactions")
```

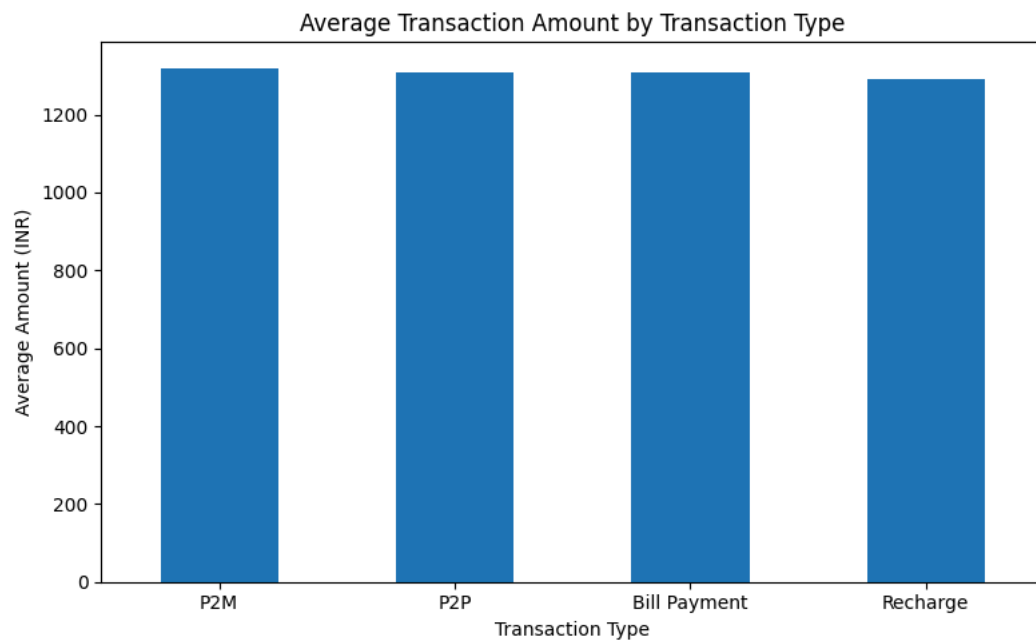
```
plt.tight_layout()
plt.show()
```



```
amount_by_type = (
    df.groupby("transaction type")["amount (INR)"]
      .mean()
      .sort_values(ascending=False)
)

plt.figure(figsize=(8, 5))
amount_by_type.plot(kind="bar")

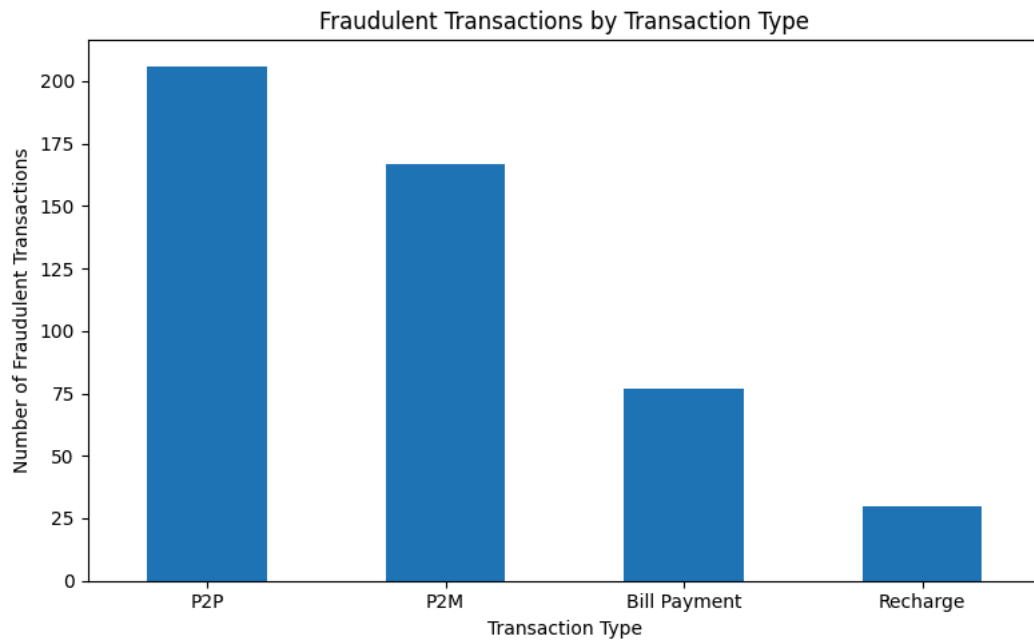
plt.title("Average Transaction Amount by Transaction Type")
plt.xlabel("Transaction Type")
plt.ylabel("Average Amount (INR)")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



```
fraud_by_type = (
    df.groupby("transaction type")["fraud_flag"]
      .sum()
      .sort_values(ascending=False)
)

plt.figure(figsize=(8, 5))
fraud_by_type.plot(kind="bar")
```

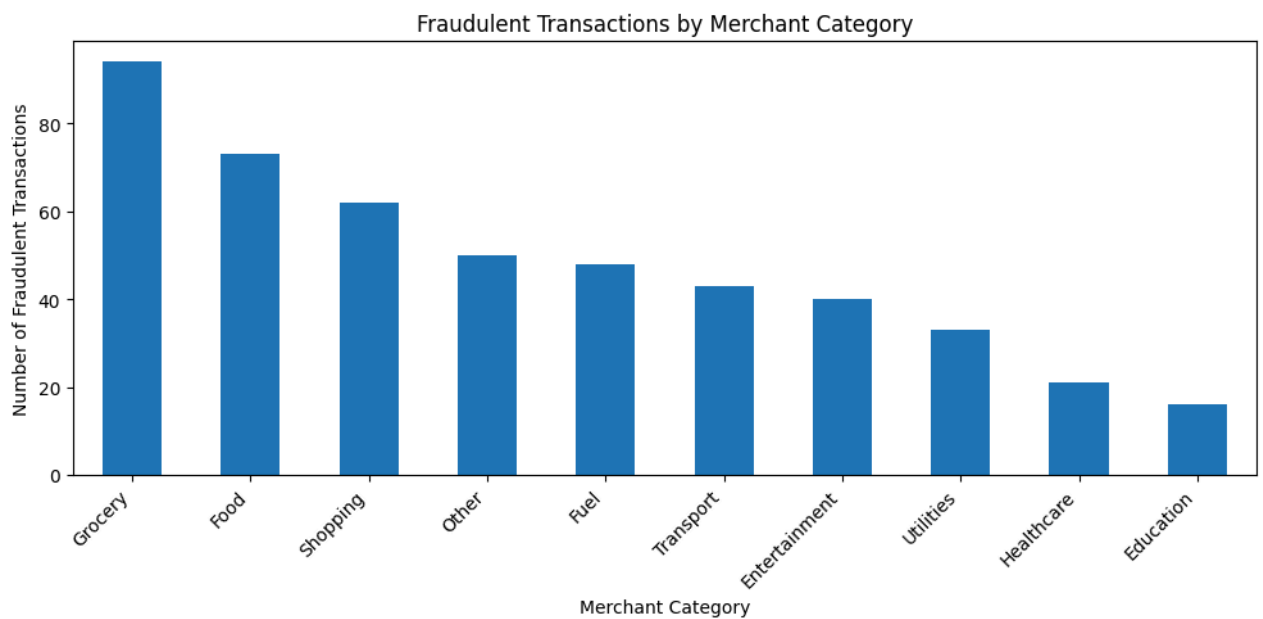
```
plt.title("Fraudulent Transactions by Transaction Type")
plt.xlabel("Transaction Type")
plt.ylabel("Number of Fraudulent Transactions")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



```
fraud_by_merchant = (
    df.groupby("merchant_category")["fraud_flag"]
      .sum()
      .sort_values(ascending=False)
)

plt.figure(figsize=(10, 5))
fraud_by_merchant.plot(kind="bar")

plt.title("Fraudulent Transactions by Merchant Category")
plt.xlabel("Merchant Category")
plt.ylabel("Number of Fraudulent Transactions")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```



```
fraud_by_hour = (
    df.groupby("hour_of_day")["fraud_flag"]
      .sum()
)
```

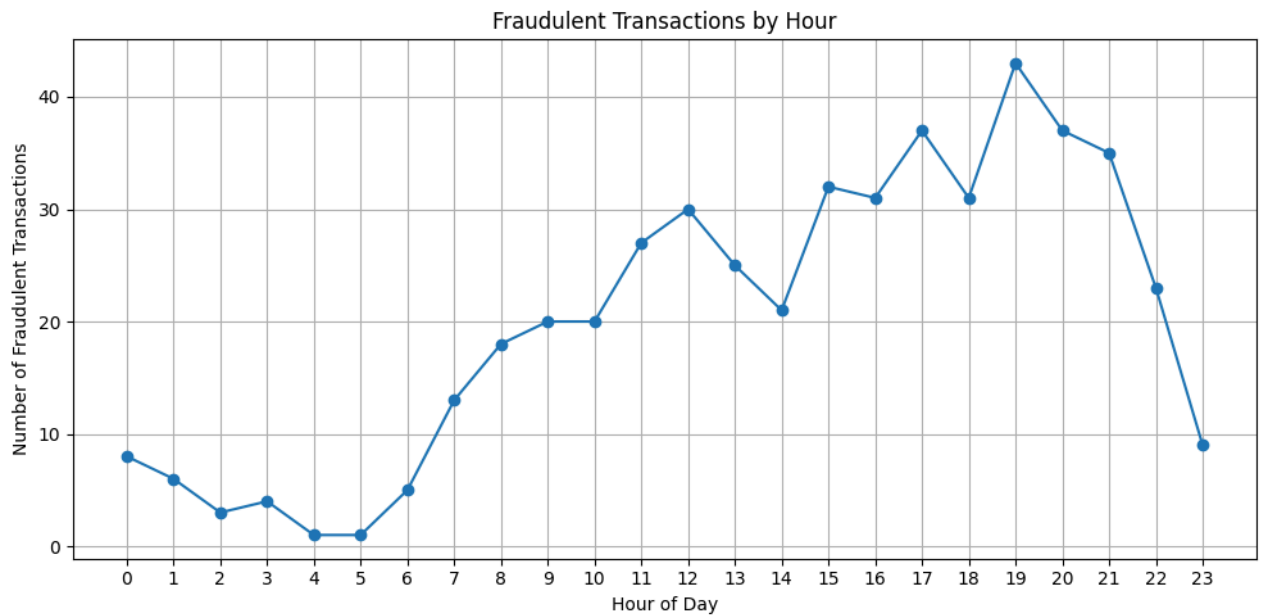
```

        .sort_index()
    )

plt.figure(figsize=(10, 5))
fraud_by_hour.plot(kind="line", marker="o")

plt.title("Fraudulent Transactions by Hour")
plt.xlabel("Hour of Day")
plt.ylabel("Number of Fraudulent Transactions")
plt.xticks(range(24))
plt.grid(True)
plt.tight_layout()
plt.show()

```



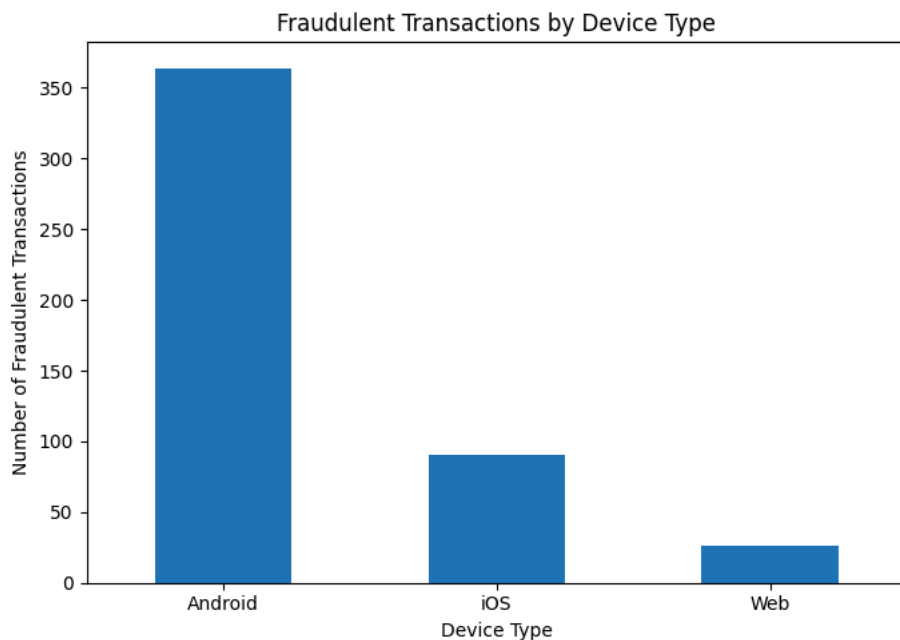
```

fraud_by_device = (
    df.groupby("device_type")["fraud_flag"]
      .sum()
      .sort_values(ascending=False)
)

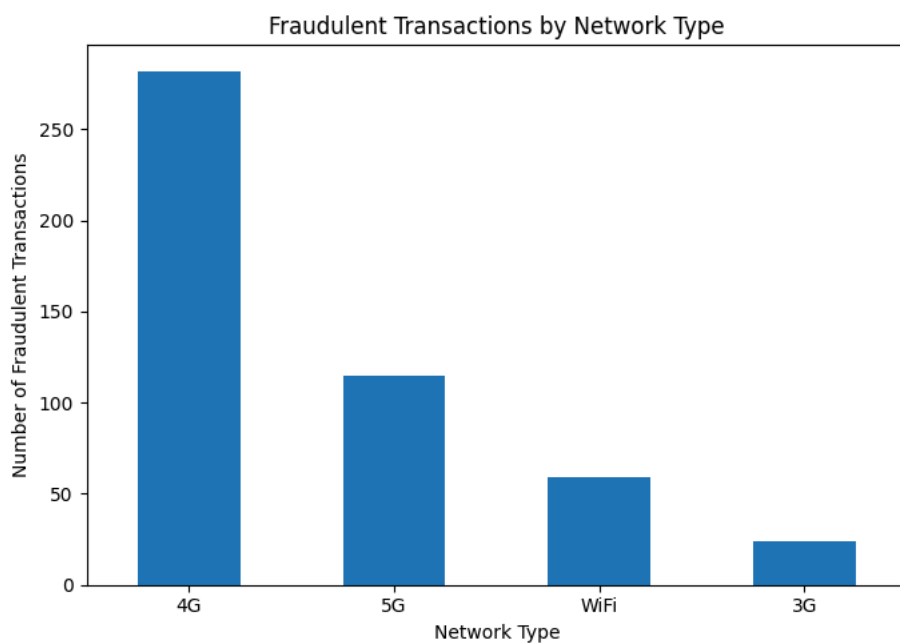
plt.figure(figsize=(7, 5))
fraud_by_device.plot(kind="bar")

plt.title("Fraudulent Transactions by Device Type")
plt.xlabel("Device Type")
plt.ylabel("Number of Fraudulent Transactions")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

```



```
fraud_by_network = (  
    df.groupby("network_type")["fraud_flag"]  
        .sum()  
        .sort_values(ascending=False)  
    )  
  
plt.figure(figsize=(7, 5))  
fraud_by_network.plot(kind="bar")  
  
plt.title("Fraudulent Transactions by Network Type")  
plt.xlabel("Network Type")  
plt.ylabel("Number of Fraudulent Transactions")  
plt.xticks(rotation=0)  
plt.tight_layout()  
plt.show()
```

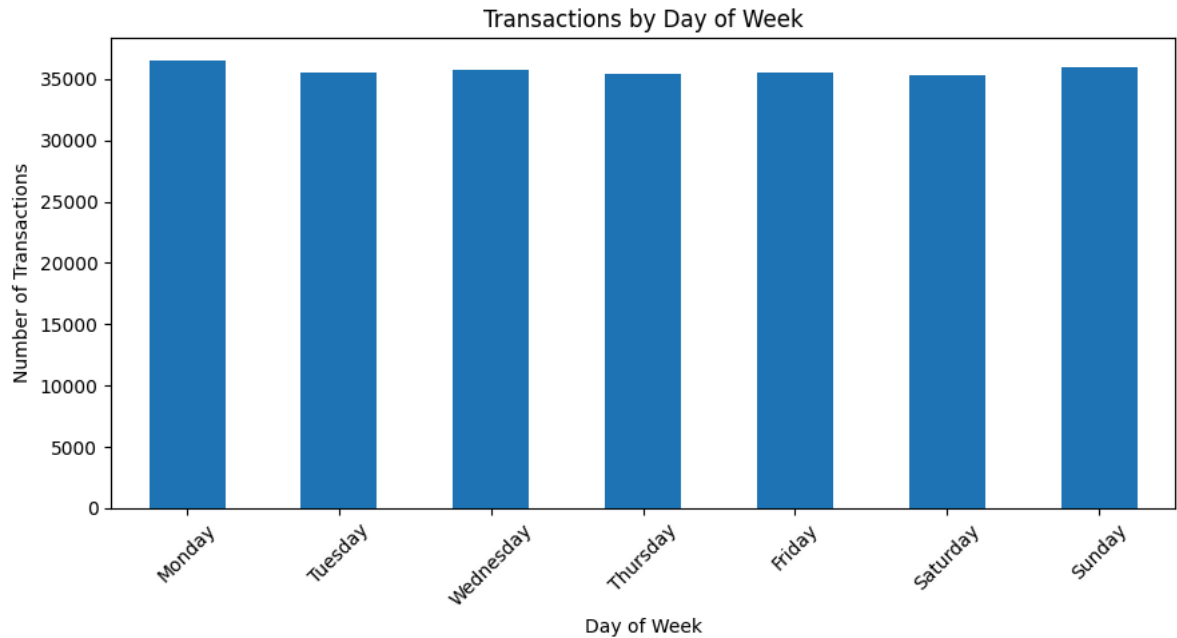


```
day_counts = df["day_of_week"].value_counts()  
  
day_order = [  
    "Monday",  
    "Tuesday",  
    "Wednesday",  
    "Thursday",  
    "Friday",  
    "Saturday",  
    "Sunday"  
]
```

```
day_counts = day_counts.reindex(day_order)

plt.figure(figsize=(9, 5))
day_counts.plot(kind="bar")

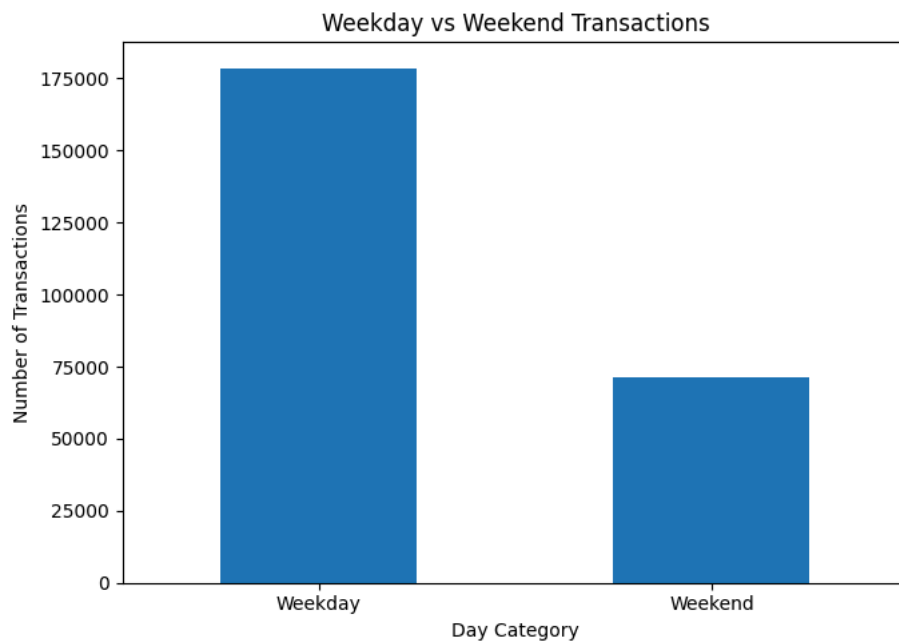
plt.title("Transactions by Day of Week")
plt.xlabel("Day of Week")
plt.ylabel("Number of Transactions")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
weekend_counts = df["is_weekend"].map({
    0: "Weekday",
    1: "Weekend"
}).value_counts()

plt.figure(figsize=(7, 5))
weekend_counts.plot(kind="bar")

plt.title("Weekday vs Weekend Transactions")
plt.xlabel("Day Category")
plt.ylabel("Number of Transactions")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



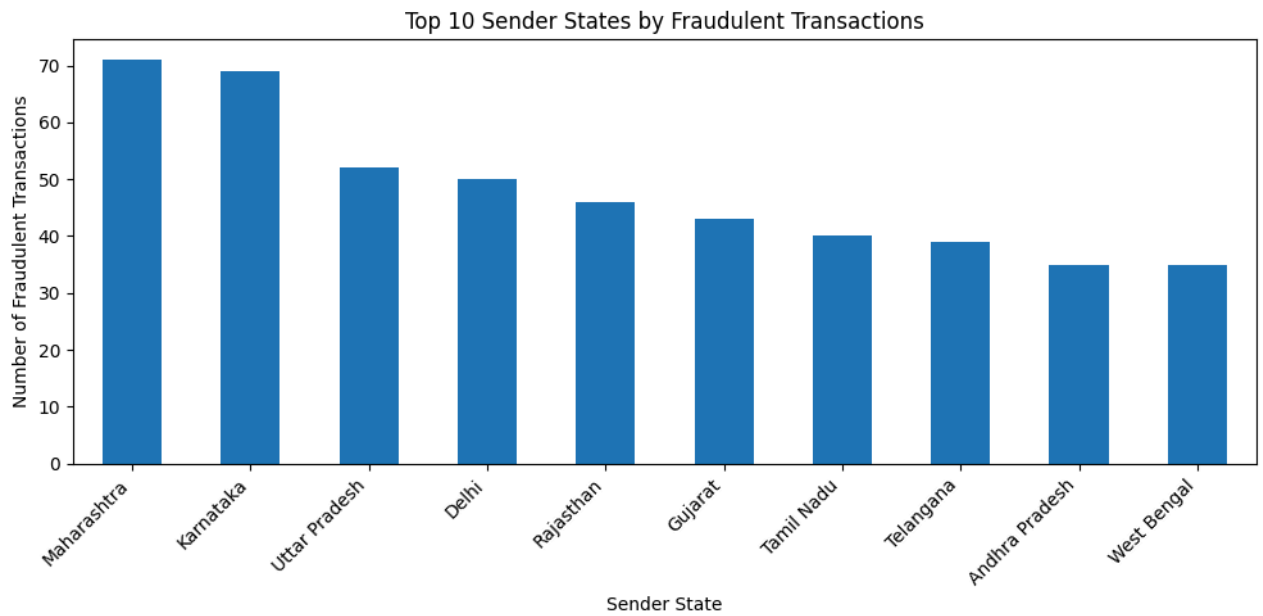
```

fraud_by_state = (
    df.groupby("sender_state")["fraud_flag"]
      .sum()
      .sort_values(ascending=False)
      .head(10)
)

plt.figure(figsize=(10, 5))
fraud_by_state.plot(kind="bar")

plt.title("Top 10 Sender States by Fraudulent Transactions")
plt.xlabel("Sender State")
plt.ylabel("Number of Fraudulent Transactions")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()

```



```

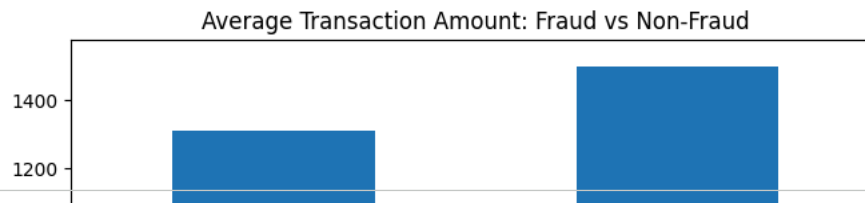
avg_amount_fraud = (
    df.groupby("fraud_flag")["amount (INR)"]
      .mean()
)

avg_amount_fraud.index = [
    "Non-Fraudulent" if x == 0 else "Fraudulent"
    for x in avg_amount_fraud.index
]

plt.figure(figsize=(7, 5))
avg_amount_fraud.plot(kind="bar")

plt.title("Average Transaction Amount: Fraud vs Non-Fraud")
plt.xlabel("Transaction Category")
plt.ylabel("Average Amount (INR)")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

```



```
print("=" * 60)
print("STAGE 7 COMPLETED")
print("=" * 60)

print("Total visualizations created : 18")

print("\nVisualizations:")
print("1. Transaction Type Distribution")
print("2. Transaction Status Distribution")
print("3. Fraud vs Non-Fraud")
print("4. Merchant Category Distribution")
print("5. Transaction Activity by Hour")
print("6. Device Type Distribution")
print("7. Network Type Distribution")
print("8. Transaction Amount Distribution")
print("9. Average Amount by Transaction Type")
print("10. Fraud by Transaction Type")
print("11. Fraud by Merchant Category")
print("12. Fraud by Hour")
print("13. Fraud by Device Type")
print("14. Fraud by Network Type")
print("15. Transactions by Day of Week")
print("16. Weekday vs Weekend")
print("17. Fraud by Sender State")
print("18. Average Amount: Fraud vs Non-Fraud")

print("\nStage 7 Data Visualization completed successfully.")
```

```
=====
STAGE 7 COMPLETED
=====
Total visualizations created : 18

Visualizations:
1. Transaction Type Distribution
2. Transaction Status Distribution
3. Fraud vs Non-Fraud
4. Merchant Category Distribution
5. Transaction Activity by Hour
6. Device Type Distribution
7. Network Type Distribution
8. Transaction Amount Distribution
9. Average Amount by Transaction Type
10. Fraud by Transaction Type
```